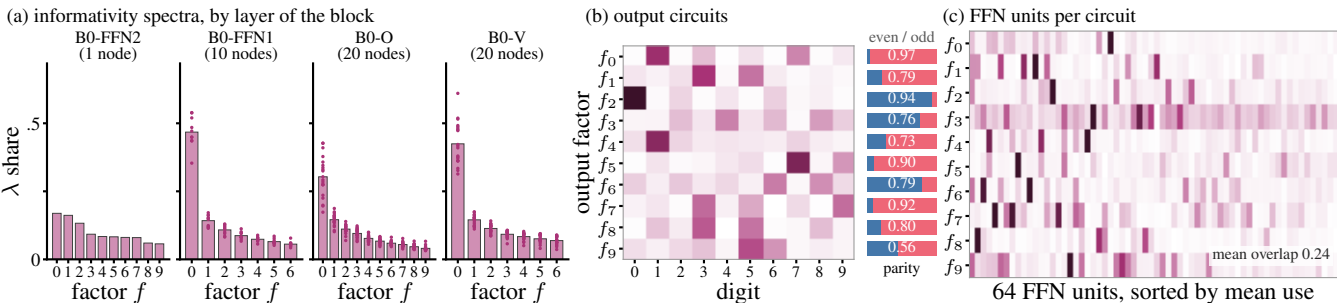


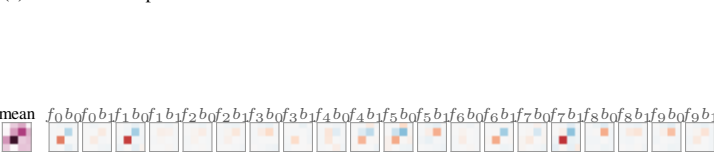


(d) L_{FFN1} sub-factors of each output circuit — weighted-average input (top, bold: dominant digit) and per-digit loading (bottom; blue even, red odd)

$1/7/1/1/3/3/1/6$ $3/2/5/5/3/2/5/6$ $0/2/0/0/0/0/6$ $4/8/4/9/4/8/2/6$ $1/7/2/1/6/1/1/6/6$ $7/8/1/1/5/3/1/6/6$ $6/8/4/9/6/8/6/6$ $9/8/4/9/4/5/9/6$ $3/8/5/1/3/3/3/6$ $5/6/5/5/5/5/5/6$



(e) CLS attention per factor



CLS-token weight 0.026; deviation of each leaf k_0 from the mean
(all 139 factor maps: pairwise cos ≥ 0.93)

(f) strongest stimuli per circuit

